KP24 Combining Classifiers: Product, Sum, Majority Vote and the Bayesian View

> Level: undergraduate, year 3
>
> Pattern Recognition | Chapter 4, §4.21

Slides: read and download

Learning objectives

- Explain the value of combining classifiers (complementary information) and write the product and sum rules with their KL derivations.
- Explain the majority-vote formula, why independence is an illusion and the three ways to create diversity.
- Explain the one-vote veto of the product rule and the Bayesian view, and know that no rule is always best.

KP24 Combining Classifiers: Product, Sum, Majority Vote and the Bayesian View

One-sentence summary: Even the best classifier errs on samples that others get right; complementary information is the whole value of combining. But combining does not guarantee a gain and there is no panacea.

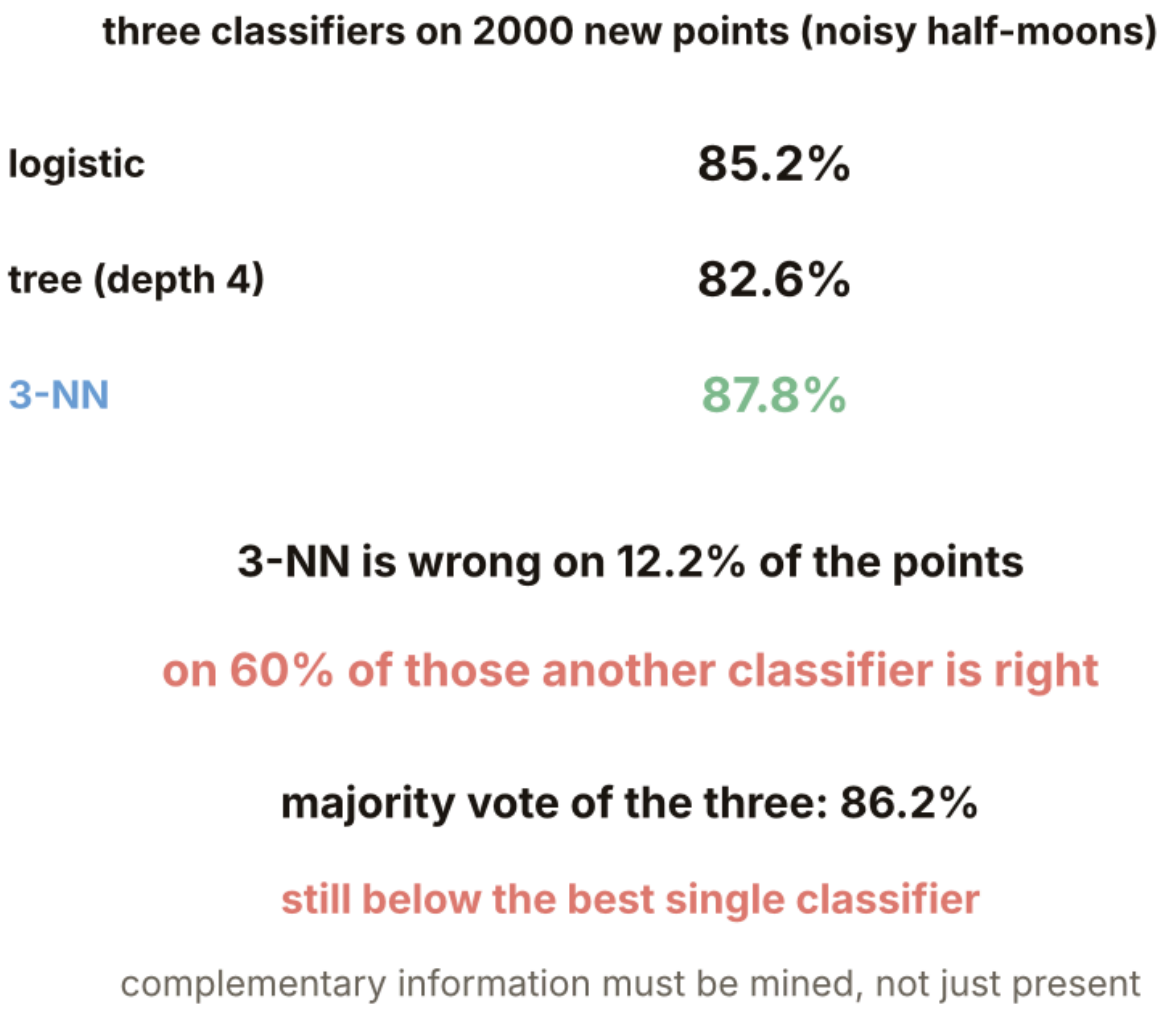
1. Complementary information

Plain explanation: What is there to gain by combining classifiers?

Example (designed here): logistic, a depth-4 tree and 3-NN on noisy half-moons (100 training points per class, 2000 new points in all): 85.2%, 82.6% and 87.8% on new points. The best, 3-NN, is wrong on 12.2%, and for 60% of those at least one other

classifier is right. But the vote of the three gets only 86.2%, below the 87.8% of 3-NN; complementary information must be mined to be worth anything.

Figure `lec05-kp24-comp`: the accuracies of three classifiers and the share of complementary errors.



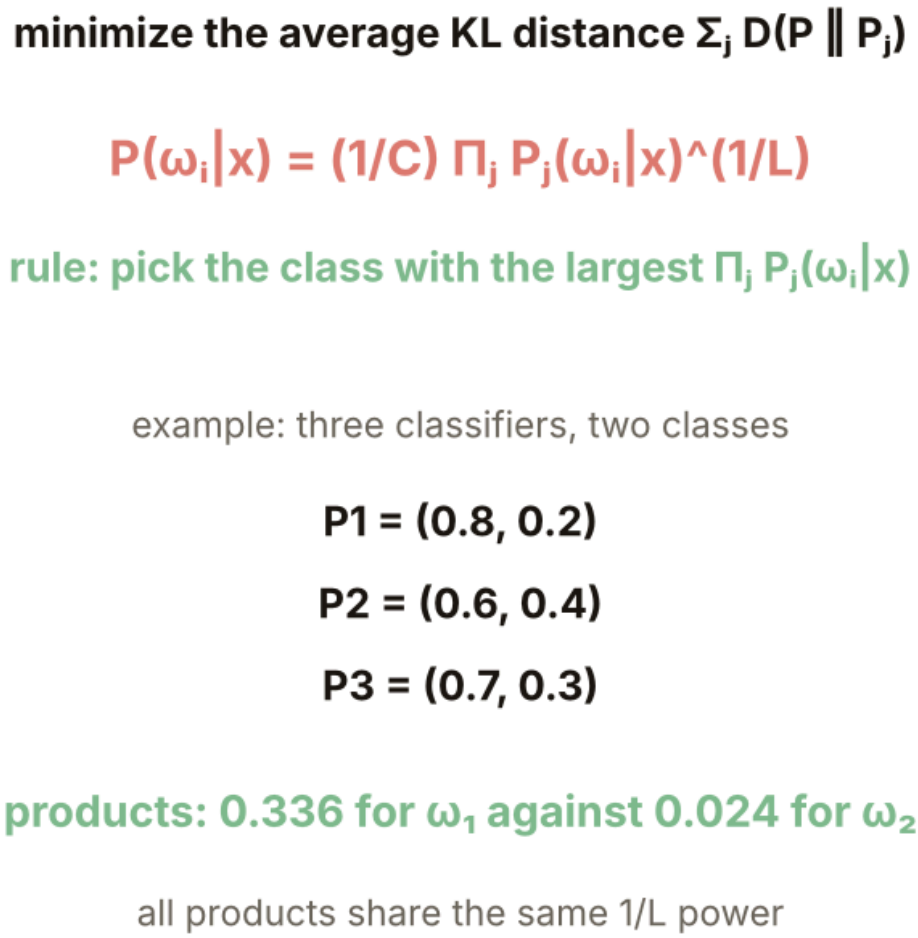
2. The product rule

KL distance $D = \sum_i P \ln(P/P_j)$: how far apart two probability distributions are; asymmetric, so strictly not a distance. The **product rule** (formal definition): minimizing $\frac{1}{L}\sum_j D_j$ gives

$$P(\omega_i|x) = \frac{1}{C}\prod_{j=1}^{L} P_j(\omega_i|x)^{1/L},$$

and the class with the largest product is chosen. **Example**: two classes, three classifiers $(0.8, 0.2)$, $(0.6, 0.4)$, $(0.7, 0.3)$: products 0.336 against 0.024, so ω_1.

Figure `lec05-kp24-prod`: the product rule and the example.



3. The sum rule

Sum rule: swap the two arguments of D_j (P_j first) and the optimum is $P(\omega_i|x) = \frac{1}{L}\sum_j P_j(\omega_i|x)$, the arithmetic mean. For the same three classifiers: means 0.70 against 0.30, also ω_1. Each rule has a rigorous information-theoretic origin; there is no theoretical preference, and in practice the product is clean but fragile when an output is near zero, and in practice the sum is often safer (source [Mill 99]).

Figure `lec05-kp24-sum`: the sum rule and the example.

swap the two arguments of the KL distance

$$D_j = \Sigma_i P_j(\omega_i|x) \ln [P_j(\omega_i|x) / P(\omega_i|x)]$$

$$P(\omega_i|x) = (1/L) \Sigma_j P_j(\omega_i|x)$$

the arithmetic mean: the sum rule

same three classifiers as the product example

sums / 3: 0.70 for ω_1 against 0.30 for ω_2

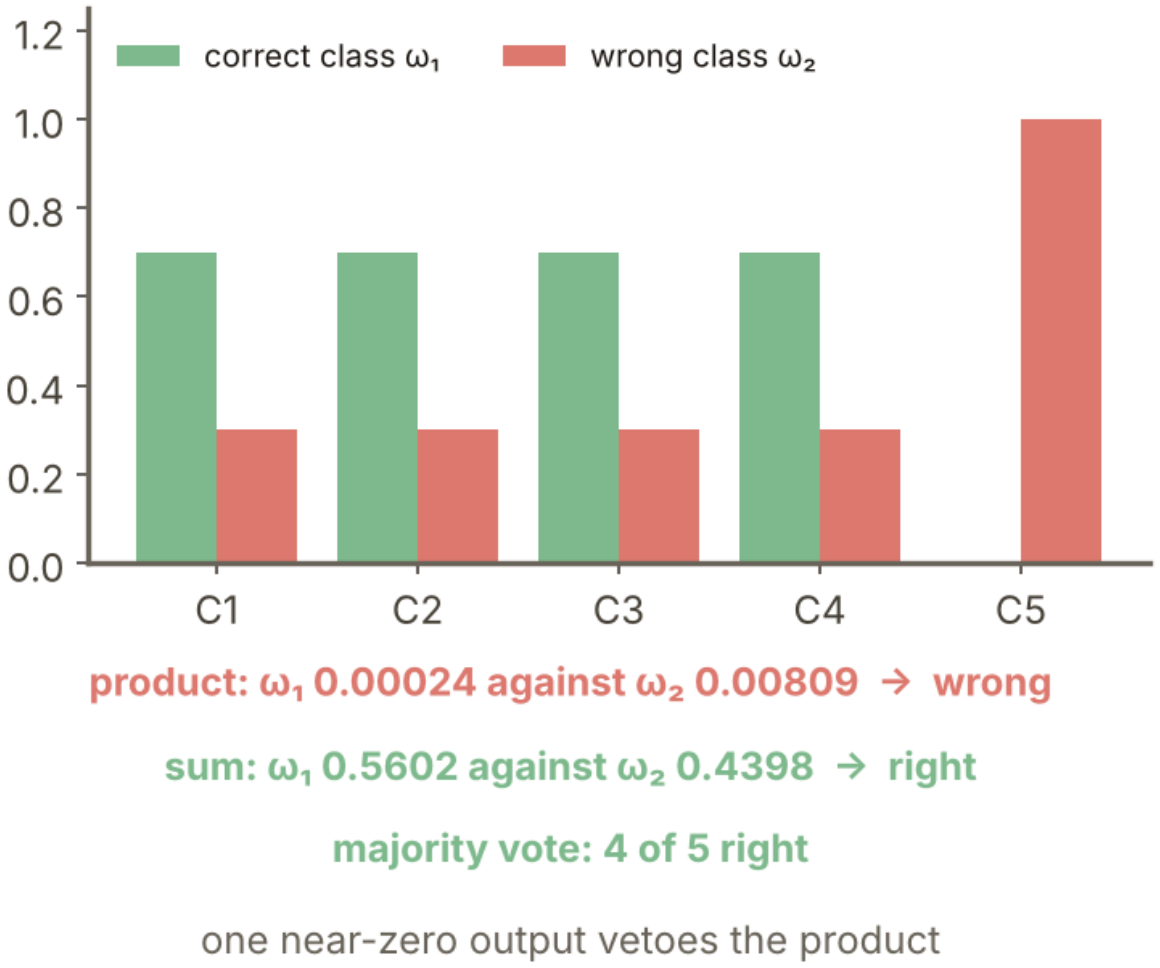
both rules choose ω_1 here

two natural rules, two different derivations

4. The veto

Example (designed here): five classifiers give the correct class ω_1 the values 0.7, 0.7, 0.7, 0.7, 0.001 (the rest to ω_2). The product gives ω_1 0.00024 against ω_2 0.00809, wrong; the sum 0.5602 against 0.4398, right; the vote is 4 to 1, right. A single near-zero factor drags the whole product to zero, a one-vote veto.

Figure `lec05-kp24-veto`: the outputs of five classifiers and the result of three rules.



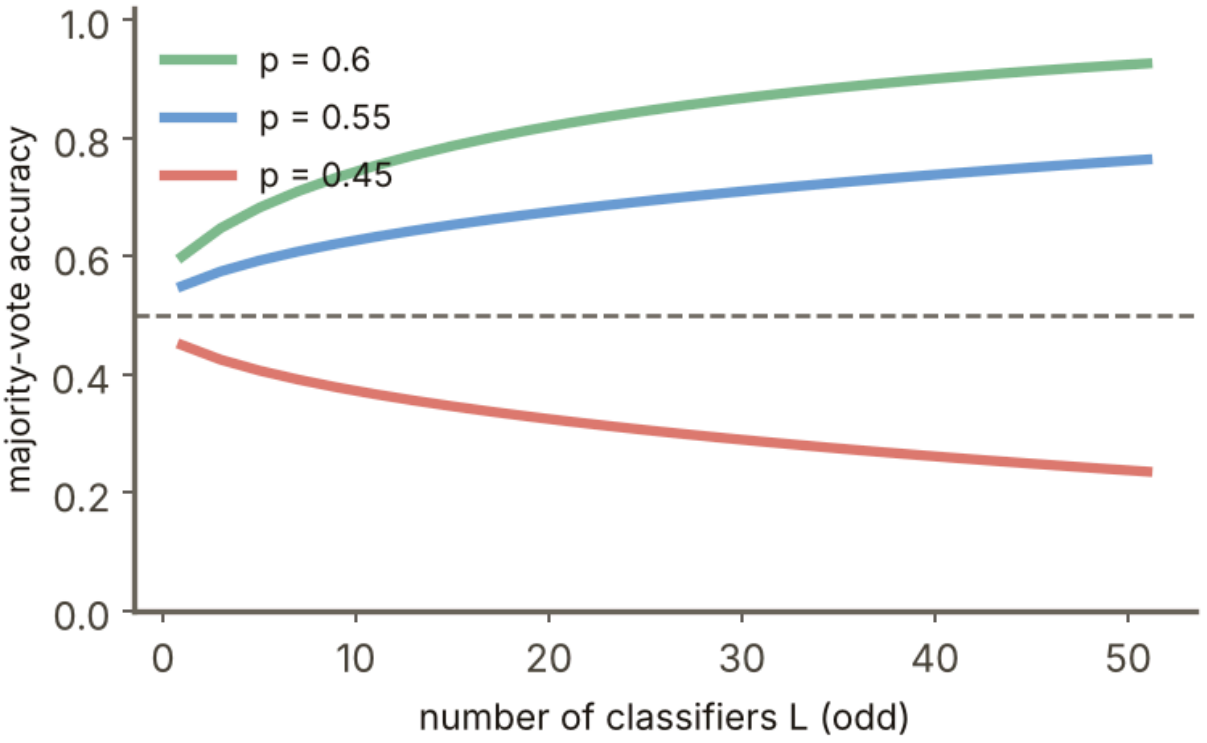
5. Majority vote

Majority vote: decide only if at least l_c classifiers agree ($l_c = \frac{L+1}{2}$ for odd L); otherwise reject. For independent decisions with equal accuracy p,

$$P_c(L) = \sum_{m=l_c}^{L} \binom{L}{m} p^m (1-p)^{L-m}.$$

$p = 0.6$: $L = 3$: 0.648, $L = 11$: 0.754, $L = 51$: 0.927; $p = 0.55$: $L = 51$: 0.764; $p = 0.45$: $L = 51$: 0.236; $p = 0.5$ stays 0.5.

Figure `lec05-kp24-vote`: $P_c(L)$ against L.



independent classifiers, each right with probability p

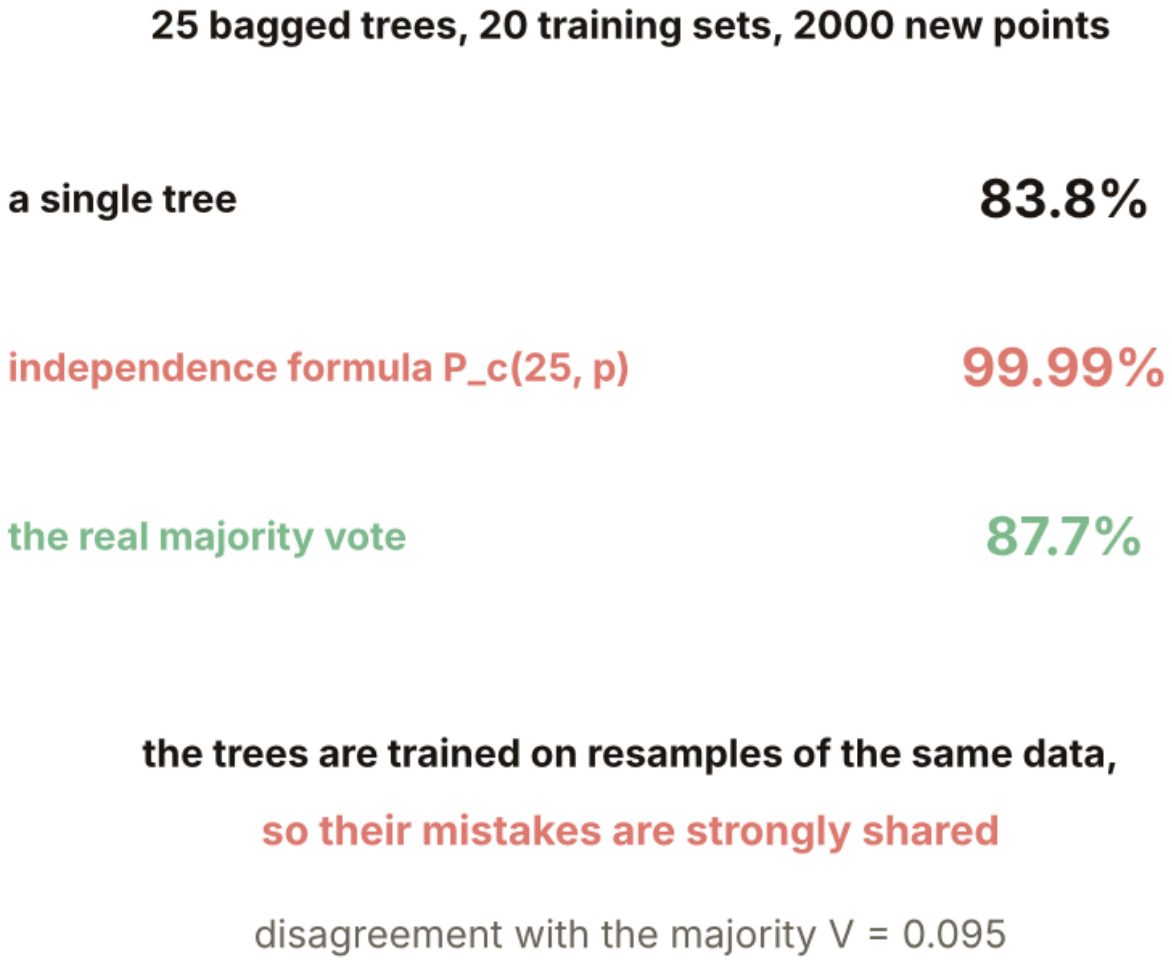
p = 0.6: 0.648 (L=3), 0.754 (L=11), 0.927 (L=51)

p = 0.45: falls towards 0; p = 0.5: stays 0.5

6. The independence illusion

Example (designed here): the half-moon data of KP23, 20 training sets, 25 bagged trees each, 2000 new points. A single tree gets 83.8% on average; the independence formula $P_c(25, p)$ gives 99.99%; the real majority vote 87.7%; the share of member decisions differing from the majority is $V = 0.095$. The trees are trained on resamples of the same data, so their mistakes are strongly shared. **Common misconception**: as long as every classifier beats guessing the vote tends to 100%; this relies on independence, which fails in practice and the more so the larger L (source).

Figure `lec05-kp24-indep`: a single tree, the independence formula and the real vote.



7. Creating diversity

Diversity: how much the members' decisions differ; one measure is V, the share of member decisions that differ from the majority. Three ways (source): the same feature space with different data points (bootstrap, most effective for unstable trees); different classifier types; different feature subspaces. **Experiment** (designed here, 10-D, 150 points, 15 runs): one tree 71.2%; 11 bootstrap trees 77.9±2.1% ($V = 0.197$); 11 random-subset trees 75.9±2.3% (0.241); 5 classifiers 82.1±1.2% (0.114). The ensemble with the largest V is not the most accurate, and the 5 classifiers include logistic regression, which happens to suit this data, so we cannot conclude that different classifiers are always better.

Figure `lec05-kp24-div`: the vote accuracy and V of three ensemble types.

10-D data, 150 training points, vote of 11 / 11 / 5 members

ensemble	vote accuracy	V
bootstrap trees	**77.9 ± 2.1**	0.197
random feature subsets	**75.9 ± 2.3**	0.241
5 different classifiers	**82.1 ± 1.2**	0.114

one tree alone: 71.2%

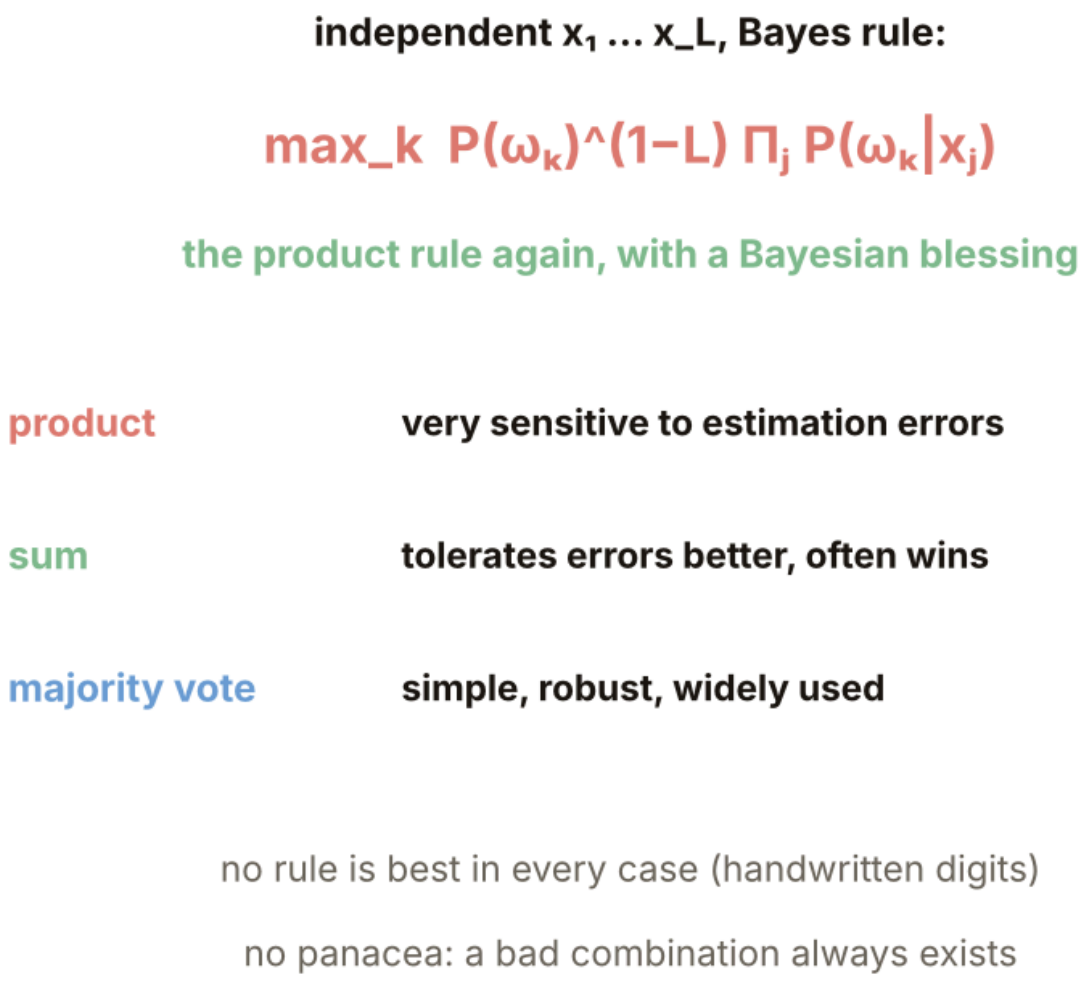
V = share of member decisions that differ from the majority

the most diverse ensemble is not the most accurate

8. The Bayesian view and how to remember

Assuming independent x_j gives $\max_k P(\omega_k)^{1-L} \prod_j P(\omega_k|x_j)$: the product rule again, but very sensitive to errors in the posterior estimates; the sum tolerates errors better and often beats the product in practice (source [Kitt 98]). The source's handwritten-digit experiments: no rule is always best; in some cases the combination is worse than the best single classifier; combining one classifier on different feature sets works best (our random subsets on page 7 did worst). Page 1 agrees: the vote gets 86.2%, below 87.8%. **To remember**: the sum and the majority vote are the most used; there is no panacea theorem.

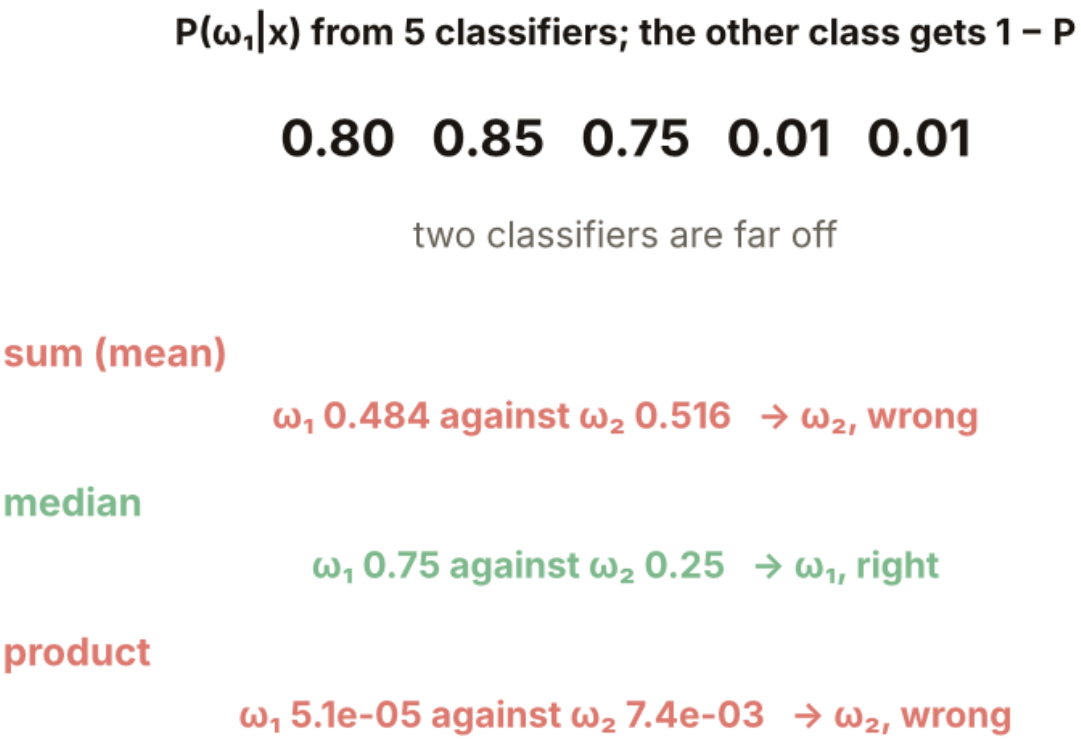
Figure `lec05-kp24-bayes`: the Bayesian view and the trade-offs of the three rules.



Extension: the median, practice and worked solution

The median (designed here): five classifiers give ω_1: 0.80, 0.85, 0.75, 0.01, 0.01. The sum gives 0.484 against 0.516, wrong; the product 5.1×10^{-5} against 7.4×10^{-3}, wrong; the median 0.75 against 0.25, right. With outlying outputs the sum is pulled away while the median holds; the source also mentions max, min and median, motivated by the chain product ≤ min ≤ mean ≤ max.

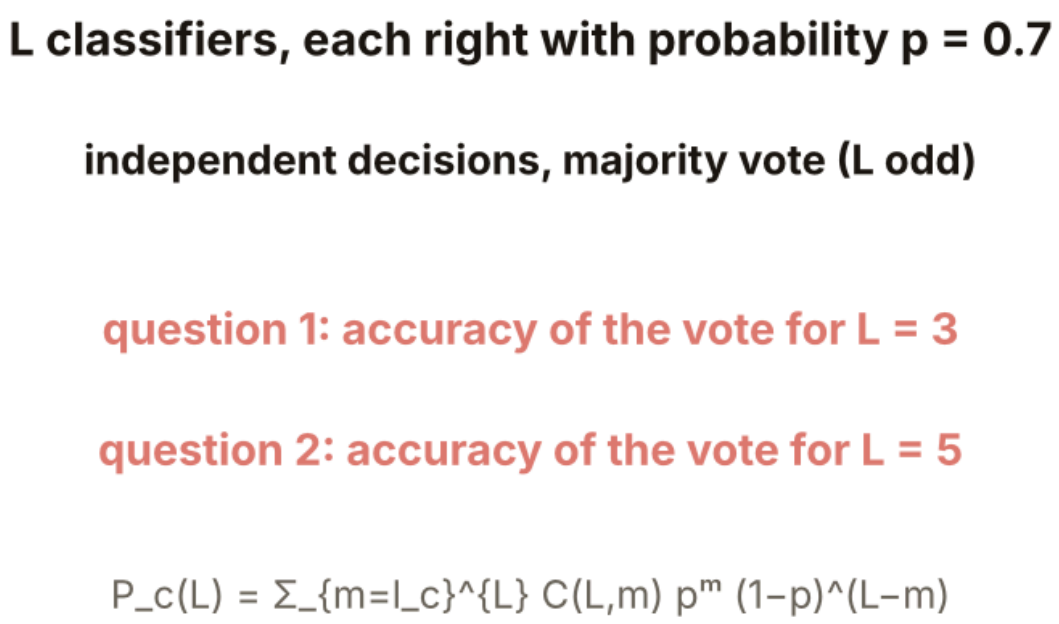
Figure `lec05-kp24-median`: the three rules with two extreme outputs.



Practice: every classifier is right with probability $p = 0.7$, decisions are independent and the majority vote has odd L.

1. What is the accuracy $P_c(3)$ for $L = 3$?
2. What is the accuracy $P_c(5)$ for $L = 5$, and how much more than for $L = 3$?

Figure `lec05-kp24-practice`: the setup of the practice.



Worked solution of questions 1 and 2: (1) $l_c = 2$: $3 \cdot 0.7^2 \cdot 0.3 + 0.7^3 = 0.441 + 0.343 = 0.784$. (2) $l_c = 3$: $10 \cdot 0.7^3 \cdot 0.3^2 + 5 \cdot 0.7^4 \cdot 0.3 + 0.7^5 = 0.3087 + 0.36015 + 0.16807 = 0.837$, 0.053 more than for $L = 3$. (3) Check: $0.7 \to 0.784 \to 0.837$ rises monotonically (under independence); real dependent classifiers do not rise this much.

Figure `lec05-kp24-practice-sol`: the computation.

$$L = 3: l_c = 2$$

$$3 \cdot 0.7^2 \cdot 0.3 + 0.7^3 = 0.441 + 0.343 = 0.784$$

$$L = 5: l_c = 3$$

$$10 \cdot 0.7^3 \cdot 0.3^2 + 5 \cdot 0.7^4 \cdot 0.3 + 0.7^5$$

$$= 0.3087 + 0.36015 + 0.16807 = 0.837$$

$$0.7 \to 0.784 \to 0.837 \text{ under independence}$$

Terms and source

中文	English
KL 距離	KL distance
乘積規則	Product rule
加總規則	Sum rule
多數投票	Majority vote
多樣性	Diversity

Source: Chapter 4, KP24, §4.21 (read 2026-10-25). Complementary information, the KL derivations of the product and sum rules, the majority vote and the independence assumption, the three ways to create diversity, the measure V, the Bayesian view, the conclusions of the Jain 00 experiments and no panacea follow the source; the three-classifier comparison, the veto example, the independence experiment, the diversity

experiment, the median example and the practice were designed and recomputed for this lecture ([added], (program-computed)).